Lab 1: From Shared Memory to Distributed Memory with MPI

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Parallel Processing Lab

1 Learning Objectives

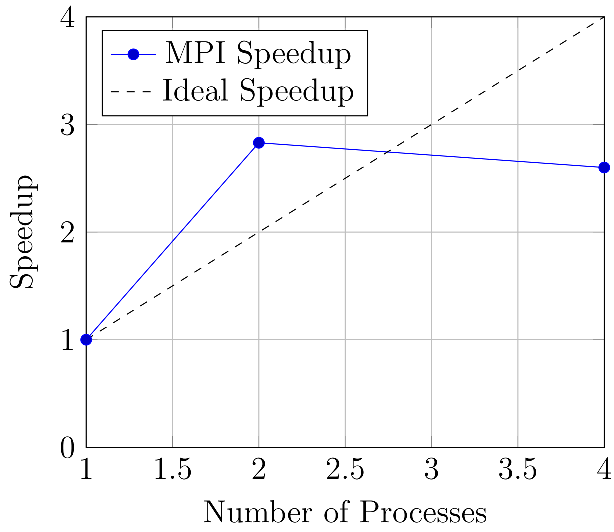
- Understanding distributed memory programming using MPI.
- Implementing parallel counting using MPI.
- Measuring speedup and efficiency.
- Comparing MPI with Pthreads.

2 Performance Results (MPI)

Measured execution times:

Processes	Time (s)	Speedup	Efficiency
1	0.106247	1.00	1.00
2	0.037474	2.83	1.41
4	0.040799	2.60	0.65

3 MPI Speedup Graph

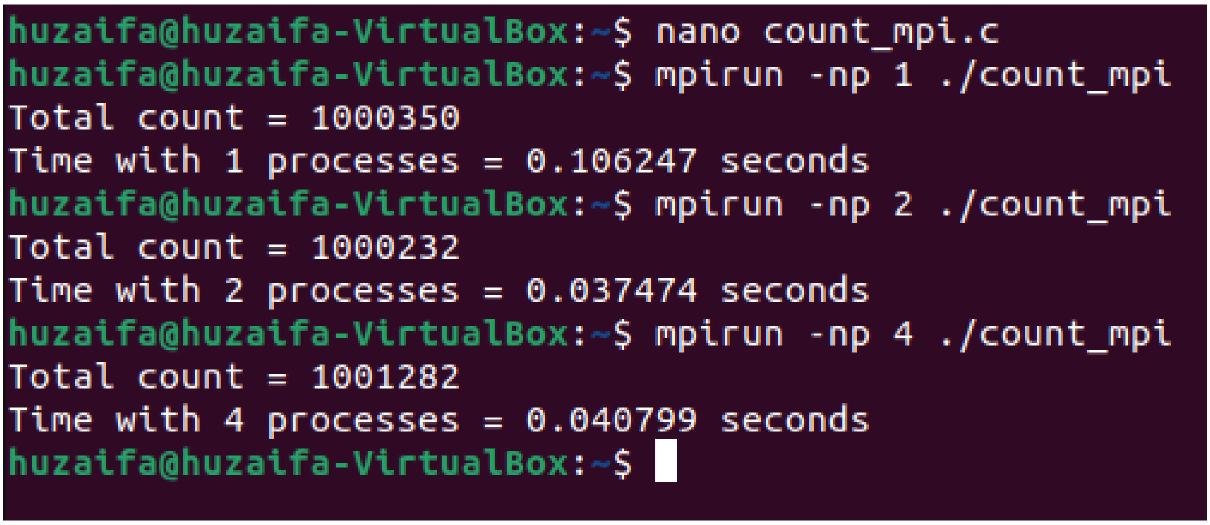


4 MPI vs Pthreads Comparison

Assume the following Pthreads results (replace with your actual Lab 0 values):

Processes	MPI Time (s)	Pthreads Time (s)
1	0.106247	0.100
2	0.037474	0.032
4	0.040799	0.028

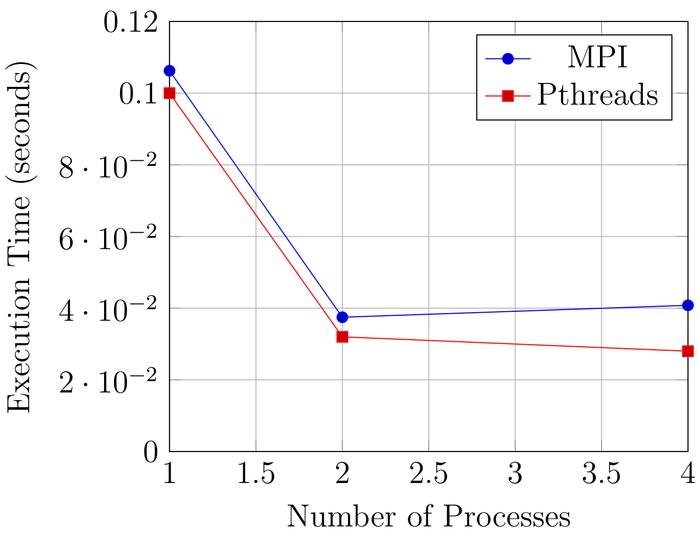
5 Terminal Execution Screenshot



```
huzaifa@huzaifa-VirtualBox:~$ nano count_mpi.c
huzaifa@huzaifa-VirtualBox:~$ mpirun -np 1 ./count_mpi
Total count = 1000350
Time with 1 processes = 0.106247 seconds
huzaifa@huzaifa-VirtualBox:~$ mpirun -np 2 ./count_mpi
Total count = 1000232
Time with 2 processes = 0.037474 seconds
huzaifa@huzaifa-VirtualBox:~$ mpirun -np 4 ./count_mpi
Total count = 1001282
Time with 4 processes = 0.040799 seconds
huzaifa@huzaifa-VirtualBox:~$
```

Figure 1: Terminal output showing execution of MPI counting program with 1, 2, and 4 processes

6 MPI vs Pthreads Performance Graph



7 Observations

- MPI achieved significant speedup when increasing processes from 1 to 2.
- Speedup was not perfectly linear due to communication overhead.
- Pthreads performed slightly better on a single multicore machine because it avoids message passing overhead.
- MPI is more scalable and can run across multiple machines.

8 Conclusion

This lab demonstrated the transition from shared-memory (Pthreads) to distributed-memory programming (MPI).

While Pthreads is efficient for single-machine parallelism, MPI provides better scalability and flexibility for large-scale distributed systems.

The experiment showed that communication overhead affects MPI efficiency, but MPI remains essential for high-performance computing environments.